

Chang Yin's

7-Day Strength Reset



Six tests. Seven short sessions. One stronger week.
Built for adults 60 to 80, chair-first, at home.

With Sun Yoon, who checks his work.

Chang Yin and Sun Yoon are AI characters created by a team of people. Their life story is fiction. The exercises and recipes are built on published research for older adults (sources in the back). This is education, not medical advice. Check with your doctor before starting new exercise.

Start here

Chang Yin

I'm Chang Yin. I'm 74. I'm also an AI character. A team of people made me, and they built this plan on published research for older adults. My life story is made up. The research is not. The sources are in the back of this book.

Here's the deal. Seven days. One short session a day, 8 to 12 minutes. Most of it next to a chair and a kitchen counter.

You won't be a new person in seven days. Nobody is. What you will have: your six numbers, written down. Seven sessions done. One habit started.

Why legs first

Strength drops faster than muscle. Around age 75, muscle mass drops about 0.6 to 1% a year, but strength drops about 2.5 to 4% a year (E04). Ten days of bed rest cost healthy adults in their late 60s about 2 pounds (0.95 kg) of leg muscle (E05).

The good part: strength answers training at every age. A review of 121 trials with 6,700 older adults training two or three times a week found large gains in strength and easier chair stands (E01). In one small study, ten nursing-home residents around age 90 trained their legs for eight weeks and their strength rose 174% (E03). Ten people is a small study. It still shows muscle adapts in the 90s.

How this week works

1. **Day 0: test.** Do the six tests (about 15 minutes) and fill in your score sheet. Work out your Strength Age.
2. **Days 1 to 7: one session a day.** Each day has a minute-by-minute plan, every move explained, with an **Easier** and a **Harder** version.
3. **One habit a day.** Seven small habits you attach to things you already do.
4. **Day 7: retest.** Same six tests. Compare.

Pick your starting level

Every move in this book has three versions. Choose for each move, not for the whole book.

If this is you	Start with
You need your hands to get out of a chair, use a walker, or feel unsteady standing	The Easier version of everything. It's chair-based. (In Strong Years this is the Rebuild level.)
You can stand up without your hands but don't exercise regularly	The main version. (Steady level.)
You exercise regularly and the main version feels easy for all the reps	The Harder version. (Strong or Iron level.)

Start easier than you think. Finish wanting one more. "Three reps. Okay, two. Okay, one, but a good one."

What you need

- A sturdy chair with no wheels, pushed against a wall.
- Your kitchen counter (a sturdy one, not a rolling island).
- A clear wall.
- A hand towel, two full water bottles, a half-gallon or gallon jug of water.
- The bottom stair with a handrail, or a sturdy low step (Day 5).
- A timer or phone, a ruler or tape measure, and a strip of tape (for the tests).
- Shoes with a grippy sole.

Before you start

Most healthy adults can begin light-to-moderate exercise like this and build up gradually (E43). Some people should talk to their doctor first.

Get your doctor's okay first if any of these are true:

- You get chest pain, pressure or tightness when you're active, or unusual shortness of breath.
- You have fainted, nearly fainted, or get dizzy when you stand up.
- You have a heart, kidney, lung or metabolic condition (like diabetes) and you have symptoms, or your doctor has told you to limit activity.
- You've had surgery, a fracture, a fall with injury, or a hospital stay in the last few months.
- You have a new hip or knee: follow your surgeon's or physical therapist's precautions first.

Set up a safe space. Chair with no wheels, pushed against a wall. Kitchen counter within reach for anything standing. Good light. No rugs under your feet. Wear shoes with a grippy sole: no socks on hard floors, no bare feet on slippery floors. Water nearby. If you can, do the tests with someone home.

Stand up slowly. Many older adults get a blood-pressure dip when they stand up quickly (about 22% of community-dwelling older people, E46). Sit a moment before you stand, then stand slowly while holding something steady. Pump your ankles ten times first.

Breathe out when it's hard. Never hold your breath to push. Breath-holding while straining spikes blood pressure (E42).

The pain rule (E41). Mild discomfort up to 3 out of 10 is okay if it settles by tomorrow. Sharper, or worse the next day: back off and ask your doctor or physical therapist.

Not for exercise. For your doctor, today: chest pain or pressure; fainting; sudden weakness or numbness in the face, arm or leg; trouble speaking; a sudden severe headache; new confusion; one calf that is swollen, red and painful; a hot, swollen joint; night pain that wakes you and doesn't change with position; loss of bladder or bowel control or numbness in the saddle area; a fall where you hit your head (especially on blood thinners). **Chest pain, face drooping, arm weakness or trouble speaking? Call 911 now.**

Your six tests

Six simple tests. About 15 minutes. Do them before Day 1, then again after Day 7, the same way, in the same shoes, at about the same time of day.

These are the same tests used in senior fitness research: the Senior Fitness Test by Rikli and Jones (normal ranges from 7,183 US adults aged 60 to 94, E49) and the CDC's STEADI screening tools for older adults (E11, E50, E51). They compare you with typical people your age. **They are fitness checks, not a medical diagnosis.** A low number is a starting line, not a verdict.

What you need: sturdy armless chair (about 17 inches high) pushed against a wall; a kitchen counter; a timer or phone; a 5-pound dumbbell (women) or 8-pound dumbbell (men), or a half-gallon jug of water (about 4.4 pounds) or a gallon jug (about 8.3 pounds); a ruler or tape measure; a strip of tape; 10 feet of clear floor. A helper makes it easier and safer.

Skip the standing tests today if you ticked anything in the Before you start list, use a walker or wheelchair, or feel unwell. Do the arm curl and the sit-and-reach seated, follow every Easier version this week (that's our chair-based Rebuild level), and show this plan to your doctor.

Test 1: 30-second chair stand

Why it matters: getting out of a chair is leg strength you use every day. The CDC uses this test in its older-adult screening (E11). Taking longer than 15 seconds for five stands is one of the European screening signs of low muscle strength (E06).

1. Chair against the wall. Sit in the middle of the seat, feet flat, about shoulder-width apart.
2. Cross your arms over your chest, hands on opposite shoulders.
3. On "Go", stand all the way up, then sit all the way down. That's one.
4. Repeat as many times as you comfortably can in 30 seconds. If you're more than halfway up at 30 seconds, it counts.
5. Stop anytime if anything hurts, you feel dizzy or short of breath.

Breathe out as you stand. If you need your hands to stand, that's okay: count your stands and write an **H** next to the number. (Officially the CDC scores it 0 when arms are used, E11. We track your number anyway so you can see progress.)

Your score: number of full stands in 30 seconds.

Typical range (middle 50% for your age, E49):

Age	Women	Men
60-64	12-17	14-19
65-69	11-16	12-18
70-74	10-15	12-17
75-79	10-15	11-17

Age	Women	Men
80–84	9–14	10–15
85–89	8–13	8–14
90–94	4–11	7–12

The CDC counts scores **below** these numbers as below average: women 60–64 under 12, 65–69 under 11, 70–79 under 10, 80–84 under 9, 85–89 under 8, 90–94 under 4; men 60–64 under 14, 65–74 under 12, 75–79 under 11, 80–84 under 10, 85–89 under 8, 90–94 under 7 (E11).

Test 2: Four-stage balance

Why it matters: the CDC uses the heel-to-toe stand as a key checkpoint: under 10 seconds is worth mentioning to your doctor (E50). In a study of 1,702 adults aged 51 to 75, people who couldn't stand on one leg for 10 seconds had higher death rates over about seven years (84% higher, E09). That is a warning sign, not a sentence. Balance is one of the most trainable things there is: practise it and it gets better, which is why it's in this week.

Stand next to your kitchen counter, one hand hovering just above it. Eyes open. You can move your arms or body to balance, but not your feet. If you touch the counter or move your feet, that stage is over. A helper stands beside you.

1. **Feet side by side**, touching. Hold 10 seconds.
2. **Half step:** the arch of one foot beside the big toe of the other. Hold 10 seconds.
3. **Heel to toe:** one foot directly in front of the other. Hold 10 seconds.

4. **One foot:** stand on one foot. Hold 10 seconds.

Stop at the first position you can't hold for 10 seconds.

Your score: the last position you held for the full 10 seconds (0 to 4), plus how many seconds you held the next one.

Test 3: 30-second arm curl

Why it matters: arm strength for carrying, lifting and pushing yourself up (E49).

1. Sit in the chair against the wall, back straight, feet flat.
2. Hold the weight in your stronger hand, arm hanging straight down beside the chair, palm facing in.
3. On "Go", curl the weight up, turning your palm up as it rises, then lower it all the way until your arm is straight.
4. Repeat as many full curls as you can in 30 seconds. Elbow stays by your side.

Breathe out as you curl. The research used 5 pounds for women and 8 pounds for men. A half-gallon jug is close to 5 pounds and a gallon jug is about 8 pounds but awkward to hold, so your number may differ from the chart. **Use the same object every time you retest.**

Your score: full curls in 30 seconds. Write down what you lifted.

Age	Women	Men
60-64	13-19	16-22
65-69	12-18	15-21
70-74	12-17	14-21

Age	Women	Men
75-79	11-17	13-19
80-84	10-16	13-19
85-89	10-15	11-17
90-94	8-13	10-14

Test 4: Two-minute step test

Why it matters: it measures your stamina for walking, stairs and chores (E49).

This one gets your heart rate up. Skip it if you have any of the "doctor first" items, if you get breathless easily, or if you have a heart condition, until your doctor says it's okay. You can slow down or rest at any time; the clock keeps running. **Stop if you feel chest pain, dizziness or unusual breathlessness.**

1. Find the height: halfway between your kneecap and the top of your hip bone. Put a strip of tape on the wall or counter at that height.
2. Stand side-on to the counter, one hand resting on it.
3. On "Go", march in place, lifting each knee up to the tape mark.
4. Count every time your **right** knee reaches the mark. March for two minutes.
5. Afterward, keep walking slowly around the kitchen for a minute. Don't sit down straight away.

Your score: number of right-knee lifts in two minutes.

Age	Women	Men
60-64	75-107	87-115
65-69	73-107	86-116
70-74	68-101	80-110
75-79	68-100	73-109
80-84	60-91	71-103
85-89	55-85	59-91
90-94	44-72	52-86

Test 5: Chair sit-and-reach (long-back version)

Why it matters: flexibility behind the legs, which you use for putting on shoes, getting in the car and bending safely (E49).

Osteoporosis, low bone density, or a past spine fracture? Skip this test. Rounding forward is the one bending pattern we avoid for fragile bones (E40). You'll track your hips and hamstrings with the stretches instead.

1. Sit on the front edge of the chair (chair against the wall). Bend one knee, foot flat.
2. Straighten your other leg in front of you, heel on the floor, toes pointing up.
3. Put one hand on top of the other, middle fingers even.

4. **Keep your back long** and hinge forward from your hips, reaching toward your toes. Go slowly. Hold two seconds. No bouncing.
5. A helper measures from your middle fingertips to the toe of your shoe. Short of the toe is a minus number (–3 inches). Past the toe is a plus number (+2 inches). Touching the toe is 0.
6. Two tries. Keep the best.

Breathe out as you reach. Because we ask you to keep your back long, which is safer, your number may look a little lower than the chart. You're comparing with yourself.

Your score: best reach in inches (plus or minus).

Age	Women	Men
60–64	–0.6 to +4.8 in	–2.5 to +4.0 in
65–69	–0.5 to +4.5 in	–3.0 to +3.0 in
70–74	–1.0 to +4.0 in	–3.5 to +2.5 in
75–79	–1.5 to +3.5 in	–4.0 to +2.0 in
80–84	–2.0 to +3.0 in	–5.5 to +1.5 in
85–89	–2.5 to +2.5 in	–5.5 to +0.5 in
90–94	–4.5 to +1.0 in	–6.5 to –0.5 in

Test 6: Timed Up and Go

Why it matters: it puts together standing up, walking, turning and sitting down, the moves behind getting around your home. The CDC uses it as a screening check: **12 seconds or longer** is worth mentioning to your doctor (E51).

1. Put a strip of tape on the floor 10 feet from the front of the chair. Clear the path.
2. Sit in the chair, back against the backrest. Wear your normal shoes; use your cane or walker if you normally do.
3. On "Go", stand up, walk to the tape **at your normal pace**, turn around, walk back and sit down.
4. The helper times you from "Go" until you are sitting. The helper walks beside you.

Your score: seconds. 12 seconds or more? Tell your doctor at your next visit. Standing up, walking and turning are exactly what this week trains, and the number usually moves with practice.

Your score sheet

Write your numbers on Day 0. Fill in the typical range for your age and sex from the charts. Then circle where you are: **Below**, **In**, or **Above** the typical range. On Day 7, write your retest number beside it.

Name: _____ Age: _____ Woman / Man

Date (Day 0): _____ Date (Day 7):

Test	Day 0	Typical for me	Below / In / Above	Day 7	Change
1. Chair stands in 30 s (H if hands used)			B I A		
2. Balance: last stage held 10 s (0-4) + seconds on the next		Stage 3 or 4	B I A		
3. Arm curls in 30 s (what I lifted: _____)			B I A		
4. Step test: right-knee lifts in 2 min			B I A		
5. Sit-and-reach, inches (+/-)			B I A		
6. Timed Up and Go, seconds		Under 12	Under 12 / 12+		

Your Strength Age

Your birthday gives you one age. Your legs and your balance give you another. Strength Age compares your chair stands and your balance with typical results for people your age and sex. It's a motivational estimate from published fitness norms, **not a medical test**. The Strong Years app adds a few questions about daily life, so the app's number can differ from this paper version by a year or two.

Step 1: chair points. Find your midpoint below (the middle of the typical range). Under 60? Use 60–64.

Age	Women midpoint	Men midpoint
60–64	14.5	16.5
65–69	13.5	15.0
70–74	12.5	14.5
75–79	12.5	14.0
80–84	11.5	12.5
85–89	10.5	11.0
90–94	7.5	9.5

Chair points = $2.5 \times (\text{midpoint} - \text{your stands})$. Round to one decimal. Keep it between –12 and +12. **If you used your hands**, chair points are at least +8.

Step 2: balance points.

Last position you held for 10 seconds	Balance points
Couldn't hold position 1	+10
Held position 1 only (feet together)	+8
Held position 2 (half step)	+5
Held position 3 (heel to toe)	0 (age 80+: -2)
Held position 4 (one foot)	-3 (age 80+: -5)

Step 3: Strength Age = your age + chair points + balance points. Round to a whole year. It can't be more than 15 years below or 20 years above your age, and never below 40.

Two worked examples.

Mary, 71. Midpoint for women 70–74 is 12.5. She did 11 stands: $2.5 \times (12.5 - 11) = +3.8$. She held heel-to-toe (position 3): **0**. $71 + 3.8 + 0 = 74.8$, so her Strength Age is **75**. Her plan: legs first, every day.

Tom, 68. Midpoint for men 65–69 is 15.0. He did 17 stands: $2.5 \times (15 - 17) = -5$. He held one foot (position 4): **-3**. $68 - 5 - 3 = 60$. His plan: keep going, try the Harder versions.

(Mary and Tom are made-up examples to show the math.)

	Day 0	Day 7
My midpoint		

	Day 0	Day 7
Chair points		
Balance points		
My Strength Age		

Day 1: Legs first

Time: about 11 minutes. **You need:** Sturdy chair against a wall, kitchen counter, a clear wall.

Chang Yin

Stand up. Sit down. That's the whole secret, done well.

Today we build the one movement you use forty times a day. Getting out of a chair. Slow and clean. Legs first.

Stop rule. Stop if you feel chest pain or pressure, dizziness, unusual shortness of breath, or sharp pain. Chest pain, face drooping, arm weakness or trouble speaking? Call 911 now. Pain rule: Mild discomfort up to 3 out of 10 is okay if it settles by tomorrow. Sharper, or worse the next day: back off and ask your doctor or physical therapist.

At a glance

Minutes	Part	Moves
0:00–2:00	Warm-up	Ankle pumps and circles (10 pumps + 5 circles each way, seated); Seated march (45 seconds); Shoulder rolls (8 slow rolls)

Minutes	Part	Moves
2:00–9:00	Main block: 2 rounds	Chair sit-to-stand (2 sets of 8 reps. Rest 45 seconds between sets); Heel raises at the counter (2 sets of 10 reps); High wall sit (2 holds of 15 seconds. Rest 30 seconds between); Standing march at the counter (45 seconds)
9:00–11:00	Cool-down	Calf stretch at the wall (20 seconds each leg, twice); Cyclic sighing (1 minute)

Warm-up (0:00–2:00)

Ankle pumps and circles

10 pumps + 5 circles each way, seated

You need: Chair or bed. **Set up:** Sit tall in your chair (or lie on your bed).

1. Point your toes away, then pull them toward you. That's one pump.
2. Then draw slow circles with your feet, both directions.

Chang's cues: Big, slow movements. Feel your calves work. **Breathe:** Breathe slowly.

Easier: Smaller movements.

Harder: Lift one foot off the floor while you circle.

Safety: Great before you stand up from sitting or lying. Sit a moment before you stand, then stand slowly while holding something steady.

Seated march

45 seconds

You need: Sturdy chair against a wall. **Set up:** Sturdy chair with no wheels and no arms (unless the easier version uses arms), pushed back against a wall so it cannot slide. Sit tall near the front of the seat, hands on the sides of the seat.

1. Lift your right knee a few inches, lower it.
2. Lift your left knee, lower it.
3. Keep a steady rhythm.

Chang's cues: Sit tall. Don't lean back. Lift from the hip. **Breathe:** Breathe steadily. Count out loud.

Easier: Lift the feet only an inch, or march more slowly.

Harder: March faster, or pump your arms too.

Shoulder rolls

8 slow rolls

You need: Chair. **Set up:** Sit or stand tall.

1. Lift your shoulders up toward your ears.
2. Roll them back and down.
3. Relax. Repeat.

Chang's cues: Big, slow circles, backward. Let your arms hang heavy.

Breathe: Breathe in as you lift, out as you roll down.

Easier: Just shrug up and let go.

Harder: Add arm circles: small circles out to the sides, forward and back.

Main block: 2 rounds (2:00–9:00)

Chair sit-to-stand

2 sets of 8 reps. Rest 45 seconds between sets.

You need: Sturdy chair against a wall. **Set up:** Sturdy chair with no wheels and no arms (unless the easier version uses arms), pushed back against a wall so it cannot slide. Sit in the middle of the seat, feet flat, hip-width apart, a little behind your knees.

1. Cross your arms over your chest, or rest your hands on your thighs.
2. Lean your chest forward, nose over toes.
3. Push the floor away and stand all the way up. Squeeze your buttocks at the top.
4. Reach your hips back and sit down slowly, like the chair is farther away than you think.

Chang's cues: Nose over toes. Knees point the same way as your toes. Don't let them cave inward. Sit down slow. Two or three seconds. **Breathe:** Breathe out as you stand. Breathe in as you sit.

Easier: Put a firm cushion or folded blanket on the seat to make it higher, or push up with your hands on your thighs or on chair arms.

Harder: Arms crossed, lower for a slow count of three, and pause one second just above the seat before touching down. Later: hold a gallon jug of water at your chest.

Heel raises at the counter

2 sets of 10 reps

You need: Kitchen counter. **Set up:** Stand facing your kitchen counter, both hands resting on it, feet hip-width apart.

1. Rise up onto the balls of your feet, as high as is comfortable.
2. Pause one second at the top.
3. Lower your heels slowly for a count of three.

Chang's cues: Tall like someone is pulling a string on top of your head. Weight over the big toe and second toe, not rolling out. **Breathe:** Breathe out as you rise.

Easier: Sit in your chair and raise your heels from there, or hold the counter with both hands and rise only halfway.

Harder: One hand on the counter. Later: one leg at a time, with both hands on the counter.

Safety: Calf cramps are common at first. Stretch and walk them out. One calf that is swollen, red and painful is not a cramp: that is for your doctor, today.

High wall sit

2 holds of 15 seconds. Rest 30 seconds between.

You need: A clear wall; chair beside you. **Set up:** Stand with your back flat against a wall, feet about 1.5 feet (a forearm's length) in front of you, hip-width apart. A chair is beside you in case you want to sit.

1. Slide your back down the wall just a few inches, a quarter of the way to sitting.
2. Hold, with your weight in your heels.
3. Slide back up to finish.

Chang's cues: Shallow is fine. This is not a contest of depth. Knees stay behind or over your toes. **Breathe:** Keep breathing the whole time. Count your breaths out loud; talking keeps you from holding your breath.

Easier: Slide down only 2 inches, or hold for just 10 seconds.

Harder: Slide a little lower (never below where your knees are bent to a right angle), or hold up to 30 seconds.

Safety: Breathe through it. High blood pressure or a heart condition? Ask your doctor before long holds.

Standing march at the counter

45 seconds

You need: Kitchen counter. **Set up:** Stand facing the counter, one or both hands resting on it.

1. Lift one knee to a comfortable height, lower it.

2. Lift the other knee.
3. Keep a steady rhythm.

Chang's cues: Stand tall. Knee up, foot down softly. **Breathe:** Breathe steadily. You should be able to talk.

Easier: Sit and do the seated march.

Harder: Lift knees higher, march faster, or let one hand hover above the counter.

Cool-down (9:00–11:00)

Calf stretch at the wall

20 seconds each leg, twice

You need: Wall or counter. **Set up:** Stand facing a wall or counter, hands on it. Step one foot back, heel down.

1. Bend the front knee and lean gently toward the wall until you feel a stretch in the back calf.
2. Hold, breathing slowly. Switch legs.

Chang's cues: Back heel stays down. Toes point forward. **Breathe:** Breathe slowly through the stretch.

Easier: Smaller step back.

Harder: Bigger step back, or bend the back knee slightly to reach the lower calf.

Cyclic sighing

1 minute

You need: Chair. **Set up:** Sit comfortably, feet flat, hands in your lap.

1. Breathe in through your nose.
2. At the top, sip in a little more air through your nose.
3. Let it all out slowly through your mouth, longer than the in-breath.
4. Repeat at a relaxed pace.

Chang's cues: Two sniffs in, one long sigh out. No holding at any point.

Breathe: The out-breath is always longer than the in-breath. No breath holds.

Easier: Just breathe slowly through your nose, making the out-breath a little longer.

Harder: Build up to 5 minutes.

Safety: If you feel light-headed, return to normal breathing.

Chang Yin

Good. Write your numbers in the tracker. Same time tomorrow.

Today's habit: Sun's protein breakfast. Put 25 to 30 grams of protein in your breakfast (E28). Two eggs plus a cup of soy milk or a small Greek yogurt gets most people there. Kidney disease? Ask your doctor for your number first.

☐ Session done ☐ Habit done Effort (1 easy to 5 hard):

____ Any pain above 3/10? ☐ No ☐ Yes

Day 2: Steady feet

Time: about 10 minutes. **You need:** Kitchen counter, sturdy chair against a wall, shoes on.

Chang Yin

Balance is a skill. Skills get better with practice.

We stay at the counter the whole time today. Hands hover. They don't wander. Hold the counter. Pride is not a safety rail.

Stop rule. Stop if you feel chest pain or pressure, dizziness, unusual shortness of breath, or sharp pain. Chest pain, face drooping, arm weakness or trouble speaking? Call 911 now. Pain rule: Mild discomfort up to 3 out of 10 is okay if it settles by tomorrow. Sharper, or worse the next day: back off and ask your doctor or physical therapist.

At a glance

Minutes	Part	Moves
0:00–1:30	Warm-up	Weight shifts (5 slow shifts each direction); Ankle pumps and circles (10 pumps, seated, then stand slowly)

Minutes	Part	Moves
1:30–8:30	Main block	Heel-to-toe stand at the counter (3 holds of 10 seconds each side); One-leg stand at the counter (3 holds of 10 seconds each side); Heel-to-toe walk along the counter (2 lengths of the counter); Side steps along the counter (2 sets of 10 steps each way); Sit-to-stand without hands (balance version) (1 set of 6 slow reps)
8:30–10:00	Cool-down	Standing hip-front stretch (20 seconds each side); 4-6 breathing (1 minute)

Warm-up (0:00–1:30)

Weight shifts

5 slow shifts each direction

You need: Kitchen counter or chair back. **Set up:** Stand facing the counter, feet hip-width apart, both hands resting on it.

1. Slowly shift your weight to your right foot. Pause.
2. Shift to your left foot. Pause.
3. Then shift forward toward your toes and back toward your heels.

Chang's cues: Move from your ankles and hips, like a tree swaying. Feet stay flat on the floor. **Breathe:** Breathe slowly.

Easier: Do it seated: shift your weight side to side on the chair.

Harder: Make the shifts bigger, or let one hand hover.

Ankle pumps and circles

10 pumps, seated, then stand slowly

Same move as before (full instructions on page 19). Cues: Big, slow movements. Feel your calves work. **Breathe:** Breathe slowly.

Safety: Great before you stand up from sitting or lying. Sit a moment before you stand, then stand slowly while holding something steady.

Easier: Smaller movements.

Harder: Lift one foot off the floor while you circle.

Main block (1:30–8:30)

Heel-to-toe stand at the counter

3 holds of 10 seconds each side

You need: Kitchen counter. **Set up:** Stand facing the counter, both hands resting on it. Shoes on.

1. Place one foot directly in front of the other, heel touching toe.
2. Stand tall and hold.
3. Switch which foot is in front.

Chang's cues: Eyes on something still at eye level. Soft knees. Tall head.

Breathe: Breathe slowly and steadily.

Easier: Half-step stance: front foot's heel beside the big toe of the back foot. Both hands on the counter.

Harder: Lift one hand to hover just above the counter. Later: both hands hovering.

Safety: Balance drills always start with support. Hands hover, they don't wander away from the counter.

One-leg stand at the counter

3 holds of 10 seconds each side

You need: Kitchen counter. **Set up:** Stand facing the counter, both hands resting on it. Shoes on.

1. Shift your weight to your left foot.
2. Lift your right foot a few inches off the floor.
3. Hold, then switch.

Chang's cues: Stand tall. Don't lean on the counter. Squeeze the buttock of the standing leg. **Breathe:** Breathe slowly. Count out loud.

Easier: Just shift your weight onto one foot, keeping the other toe lightly on the floor.

Harder: One hand hovering above the counter, then both hands hovering, right next to the counter.

Safety: No barefoot drills on slippery floors, no socks on hard floors, no rugs under your feet.

Heel-to-toe walk along the counter

2 lengths of the counter

You need: Kitchen counter. **Set up:** Stand at one end of the counter, one hand resting on it. Clear floor, shoes on.

1. Step forward, placing your heel directly in front of the toes of the other foot.
2. Take 10 steps along the counter.
3. Turn carefully and walk back.

Chang's cues: Eyes forward, not down at your feet. Slow is correct.

Breathe: Breathe steadily.

Easier: Walk with your feet slightly apart, not touching, holding the counter.

Harder: Let your hand hover above the counter. Later: walk backward along the counter with one hand resting on it (normal steps).

Side steps along the counter

2 sets of 10 steps each way

You need: Kitchen counter. **Set up:** Stand facing the counter, both hands resting on it.

1. Step sideways to the right, then bring your left foot to meet it.
2. Take 10 steps to the right, then 10 back to the left.
3. Toes point forward the whole time.

Chang's cues: Stay tall. Small, controlled steps. Feet don't cross. **Breathe:** Breathe normally. Talk or count out loud.

Easier: Fewer steps, slower.

Harder: Bend your knees slightly and stay low, or tie a resistance loop above your knees.

Sit-to-stand without hands (balance version)

1 set of 6 slow reps

You need: Sturdy chair against a wall. **Set up:** Sturdy chair with no wheels and no arms (unless the easier version uses arms), pushed back against a wall so it cannot slide.

1. Sit with feet slightly narrower than usual.
2. Stand up without using your hands.
3. Balance for two seconds at the top before sitting back down slowly.

Chang's cues: Find your balance at the top before you move. Slow down.

Breathe: Breathe out as you stand.

Easier: Use your hands on your thighs.

Harder: Feet together.

Cool-down (8:30–10:00)

Standing hip-front stretch

20 seconds each side

You need: Kitchen counter. **Set up:** Stand side-on to the counter, near hand on it. Step one foot back into a small split stance.

1. Tuck your tailbone under slightly and squeeze the buttock of the back leg.
2. Shift your hips forward a little until you feel a gentle stretch at the front of the back hip.
3. Hold, then switch sides.

Chang's cues: Stay tall. Don't arch your back. Small movement, big squeeze. **Breathe:** Breathe slowly.

Easier: Smaller step.

Harder: Reach the arm on the back-leg side up toward the ceiling, to comfortable height.

4-6 breathing

1 minute

You need: Chair. **Set up:** Sit tall, one hand on your belly.

1. Breathe in through your nose for a count of 4.
2. Breathe out gently for a count of 6.
3. No pause at the top or bottom.

Chang's cues: Belly rises first. Soft shoulders. **Breathe:** In for 4, out for 6. That is about 6 breaths a minute. No holds.

Easier: In for 3, out for 4.

Harder: Continue for 5 minutes.

Safety: If you feel light-headed, go back to normal breathing.

Chang Yin

Tell me your longest one-leg time. Honest number. Same time tomorrow.

Today's habit: Toothbrush balance. While you brush your teeth, stand heel-to-toe with your free hand on the sink or a grab bar. Switch feet halfway. Two minutes, twice a day, is 28 minutes of balance practice a week without trying. Home balance practice is a core part of the Otago programme (E13).

☐ Session done ☐ Habit done Effort (1 easy to 5 hard):

____ Any pain above 3/10? ☐ No ☐ Yes

Day 3: Arms, back and grip

Time: about 11 minutes. **You need:** Wall, kitchen counter, a hand towel, two water bottles, a half-gallon or gallon jug, chair against a wall.

Chang Yin

Grip is a number worth caring about.

In a study of 142,861 adults in 17 countries, every 5 kilograms of lower grip strength was linked with 16% higher death rates over the study (E07). A link, not a cause. But grip, pulling and carrying are how you live. Jars. Grocery bags. Grandkids.

Stop rule. Stop if you feel chest pain or pressure, dizziness, unusual shortness of breath, or sharp pain. Chest pain, face drooping, arm weakness or trouble speaking? Call 911 now. Pain rule: Mild discomfort up to 3 out of 10 is okay if it settles by tomorrow. Sharper, or worse the next day: back off and ask your doctor or physical therapist.

At a glance

Minutes	Part	Moves
0:00–2:00	Warm-up	Shoulder rolls (8 rolls); Wall slides (6 slow slides); Chin tuck (6 tucks, 2-second hold)

Minutes	Part	Moves
2:00–10:00	Main block: 2 rounds	Wall push-up (2 sets of 8); One-arm jug row at the counter (2 sets of 8 each arm); Water-bottle curl (2 sets of 10); Seated overhead press (to comfortable height) (2 sets of 8); Towel wring (grip) (3 wrings each direction, 3-second squeeze); Grocery carry (2 walks of 30 seconds)
10:00–11:00	Cool-down	Seated chest lift (6 lifts); Cyclic sighing (30 seconds)

Warm-up (0:00–2:00)

Shoulder rolls

8 rolls

Same move as before (full instructions on page 20). Cues: Big, slow circles, backward. Let your arms hang heavy. **Breathe:** Breathe in as you lift, out as you roll down.

Easier: Just shrug up and let go.

Harder: Add arm circles: small circles out to the sides, forward and back.

Wall slides

6 slow slides

You need: A clear wall. **Set up:** Stand with your back against a wall, feet a few inches out. Forearms on the wall in a goalpost shape, elbows bent.

1. Slide your arms up the wall only as far as comfortable.

2. Slide back down to the goalpost.

Chang's cues: Ribs down. Slow and smooth. **Breathe:** Breathe in as you slide up, out as you slide down.

Easier: Face the wall and walk your fingertips up it instead.

Harder: Hold for two seconds at the top.

Safety: Shoulder pain or a known rotator cuff problem? Only lift to a comfortable height, or skip it.

Chin tuck

6 tucks, 2-second hold

You need: Chair or wall. **Set up:** Sit or stand tall, back of the head near a wall if you like.

1. Slide your chin straight back, making a double chin. Keep your eyes level.
2. Hold two seconds. Relax.

Chang's cues: Slide back, don't tip down. Tall neck. **Breathe:** Breathe normally.

Easier: Smaller movement, one-second hold.

Harder: Hold five seconds.

Safety: Nods and half-turns only. Never full neck circles.

Main block: 2 rounds (2:00–10:00)

Wall push-up

2 sets of 8

You need: A clear wall. **Set up:** Stand arm's length from a wall. Hands on the wall at shoulder height, a little wider than your shoulders. Shoes on, non-slip floor.

1. Bend your elbows and bring your chest toward the wall.
2. Elbows point back at an angle, not straight out.
3. Push the wall away until your arms are straight.

Chang's cues: Body in one straight line, head to heels. Chest to the wall, not your chin. **Breathe:** Breathe out as you push away.

Easier: Stand closer to the wall and move only halfway.

Harder: Step your feet farther back. When 15 are easy, move to counter push-ups.

One-arm jug row at the counter

2 sets of 8 each arm

You need: Kitchen counter + a jug or bag (start with a half-full gallon jug, about 4 pounds). **Set up:** Stand side-on to the counter. Put your left hand on the counter. Step your left foot forward and hinge forward a little with a long, flat back. Jug in your right hand, arm hanging.

1. Pull the jug up toward your ribs, elbow close to your body.
2. Squeeze your shoulder blade toward your spine.
3. Lower slowly. Do all reps, then switch sides.

Chang's cues: Pull with your elbow, not your hand. Long back. Don't twist.

Breathe: Breathe out as you pull.

Easier: Stand upright and pull a towel looped around a sturdy door handle (see Towel row), or use a lighter jug.

Harder: Full gallon jug (about 8 pounds), or a bag with a few cans. Pause for one second at the top.

Safety: Osteoporosis: keep the hinge small and the back long.

Water-bottle curl

2 sets of 10

You need: Two full water bottles, or two soup cans; chair. **Set up:** Sturdy chair with no wheels and no arms (unless the easier version uses arms), pushed back against a wall so it cannot slide. Sit tall, a bottle in each hand, arms hanging, palms facing forward.

1. Bend your elbows and bring the bottles toward your shoulders.
2. Lower slowly all the way down.

Chang's cues: Elbows stay by your sides. No swinging. **Breathe:** Breathe out as you curl up.

Easier: One arm at a time, or empty bottles.

Harder: Heavier: a half-gallon jug (about 4 pounds), then a full gallon jug (about 8 pounds), one at a time.

Seated overhead press (to comfortable height)

2 sets of 8

You need: Two water bottles; chair. **Set up:** Sturdy chair with no wheels and no arms (unless the easier version uses arms), pushed back against a wall so it cannot slide. Sit tall, a bottle in each hand at shoulder height, palms facing each other.

1. Press the bottles up toward the ceiling, only as high as is comfortable.
2. Lower slowly back to your shoulders.

Chang's cues: Ribs down. Don't arch your back. Only to comfortable height. **Breathe:** Breathe out as you press up.

Easier: One arm at a time, or press forward and up at a slant instead of straight up.

Harder: Heavier bottles or a half-gallon jug in each hand; stand at the counter to press.

Safety: Shoulder pain or a known rotator cuff problem? Only lift to a comfortable height, or skip it.

Towel wring (grip)

3 wrings each direction, 3-second squeeze

You need: A small towel. **Set up:** Sit or stand. Hold a rolled hand towel with both hands in front of you, like a bicycle handlebar.

1. Wring the towel hard, like squeezing out water: right hand one way, left hand the other.
2. Hold three seconds. Switch directions.

Chang's cues: Squeeze from your whole hand, not just your fingertips. Shoulders relaxed. **Breathe:** Breathe out as you squeeze.

Easier: Squeeze a soft ball or rolled sock instead.

Harder: Use a damp towel, or hold each wring for five seconds.

Safety: Hand arthritis flaring today? Squeeze gently, or skip and do the finger spread only.

Grocery carry

2 walks of 30 seconds

You need: Two grocery bags, jugs or kettlebells; clear hallway. **Set up:** Hold a bag or jug in each hand at your sides. Clear path, shoes on, walk along a counter or hallway wall where possible.

1. Stand tall, shoulders down.
2. Walk slowly and steadily for the set time.
3. Set the load down by hinging at the hips with a long back, not by rounding.

Chang's cues: Tall, like you're proud of the groceries. Short, steady steps. **Breathe:** Breathe steadily. You should be able to talk.

Easier: One light bag, and keep your free hand trailing along the counter.

Harder: Heavier: two full gallon jugs (about 8 pounds each). Iron track: two heavier bags or kettlebells.

Safety: Pick up and put down with a hinge. Never round your back to lift.

Cool-down (10:00–11:00)

Seated chest lift

6 lifts

You need: Chair. **Set up:** Sit near the front of the seat, feet flat, hands resting on your thighs.

1. Lift your chest up toward the ceiling, growing tall.
2. Gently squeeze your shoulder blades back.
3. Relax without slumping.

Chang's cues: Grow tall. Look straight ahead, not up. **Breathe:** Breathe in as you lift, out as you relax.

Easier: Smaller lift.

Harder: Hold three seconds at the top.

Cyclic sighing

30 seconds

Same move as before (full instructions on page 25). **Cues:** Two sniffs in, one long sigh out. No holding at any point. **Breathe:** The out-breath is always longer than the in-breath. No breath holds.

Safety: If you feel light-headed, return to normal breathing.

Easier: Just breathe slowly through your nose, making the out-breath a little longer.

Harder: Build up to 5 minutes.

Chang Yin

Carry your groceries in two bags this week. One in each hand. Same time tomorrow.

Today's habit: Two-bag carry. Every time you bring groceries in, carry two lighter bags, one in each hand, standing tall. It's the carry exercise with a purpose. Pick bags up with a hinge, long back, never a rounded one.

☐ Session done ☐ Habit done Effort (1 easy to 5 hard):
____ Any pain above 3/10? ☐ No ☐ Yes

Day 4: Loosen up

Time: about 11 minutes. **You need:** Chair against a wall, kitchen counter, a wall.

Chang Yin

I'd rather lift than stretch. Sun knows this. Sun is standing behind me.

Mobility is how strength gets to use its full range. For back stiffness, exercise has the best evidence of anything: 249 trials, 24,486 people (E30). Nothing forced today. Long back. Easy breathing.

Stop rule. Stop if you feel chest pain or pressure, dizziness, unusual shortness of breath, or sharp pain. Chest pain, face drooping, arm weakness or trouble speaking? Call 911 now. Pain rule: Mild discomfort up to 3 out of 10 is okay if it settles by tomorrow. Sharper, or worse the next day: back off and ask your doctor or physical therapist.

At a glance

Minutes	Part	Moves
0:00–5:00	Seated	Ankle pumps and circles (10 pumps + 5 circles each way); Neck nods and half-turns (5 nods, 3 half-turns each side); Gentle seated turn (6 gentle turns each side); Seated chest lift (8 lifts); Seated hamstring stretch (long back) (2 holds of 20 seconds each leg); Seated hip stretch (figure-4) (20 seconds each side)
5:00–10:00	Standing at the counter	Standing hip-front stretch (2 holds of 20 seconds each side); Calf stretch at the wall (20 seconds each leg); Standing bird-dog at the counter (6 each side); Counter hip hinge (8 slow hinges)
10:00–11:00	Close	4-6 breathing (1 minute)

Seated (0:00–5:00)

Ankle pumps and circles

10 pumps + 5 circles each way

Same move as before (full instructions on page 19). Cues: Big, slow movements. Feel your calves work. **Breathe:** Breathe slowly.

Safety: Great before you stand up from sitting or lying. Sit a moment before you stand, then stand slowly while holding something steady.

Easier: Smaller movements.

Harder: Lift one foot off the floor while you circle.

Neck nods and half-turns

5 nods, 3 half-turns each side

You need: Chair. **Set up:** Sit tall.

1. Slowly nod yes: look a little down, then back to level.
2. Slowly turn your head to the right, only as far as comfortable. Back to center.
3. Turn to the left. Back to center.

Chang's cues: Small and slow. Chin stays level when you turn. **Breathe:** Breathe slowly.

Easier: Make the movement even smaller.

Harder: Add a chin tuck: gently slide your chin straight back, making a double chin, hold two seconds.

Safety: Nods and half-turns only. Never full neck circles. Dizzy with head movements? Keep it tiny, keep your eyes open, and tell your doctor.

Gentle seated turn

6 gentle turns each side

You need: Chair. **Set up:** Sit tall, feet flat. Cross your arms over your chest.

1. Turn your chest to the right, only as far as comfortable. Eyes follow.
2. Return to center.
3. Turn to the left.

Chang's cues: Grow tall first, then turn. Hips stay still. **Breathe:** Breathe out as you turn.

Easier: Smaller turns.

Harder: Place one hand on the opposite knee as a light guide. Never pull into the turn.

Safety: Osteoporosis? Turn gently and only as far as comfortable. Never force the twist.

Seated chest lift

8 lifts

Same move as before (full instructions on page 42). **Cues:** Grow tall. Look straight ahead, not up. **Breathe:** Breathe in as you lift, out as you relax.

Easier: Smaller lift.

Harder: Hold three seconds at the top.

Seated hamstring stretch (long back)

2 holds of 20 seconds each leg

You need: Chair. **Set up:** Sturdy chair with no wheels and no arms (unless the easier version uses arms), pushed back against a wall so it cannot slide. Sit near the front edge. Straighten one leg in front of you, heel on the floor, toes up.

1. Sit tall, hands on your bent knee.
2. Hinge forward from your hips with a long, flat back until you feel a stretch behind the straight knee.
3. Hold. Switch legs.

Chang's cues: Chest proud. Bend from the hips, not the back. A gentle stretch is enough. **Breathe:** Breathe slowly.

Easier: Lean only an inch or two.

Harder: Pull your toes toward you a little more.

Safety: Osteoporosis or a past spine fracture: keep the back long. Never curl down toward your toes.

Seated hip stretch (figure-4)

20 seconds each side

You need: Chair. **Set up:** Sturdy chair with no wheels and no arms (unless the easier version uses arms), pushed back against a wall so it cannot slide. Sit tall.

1. Rest your right ankle on your left knee, or as close as you can.
2. Sit tall and let the right knee relax down.

3. For more, hinge forward a little with a long back. Hold. Switch.

Chang's cues: Tall back. Relax the knee, don't push it. **Breathe:** Breathe slowly.

Easier: Rest the ankle on your shin instead of your knee, or skip crossing and just let the knee drop out a little.

Harder: Hinge forward a little farther with a long back.

Safety: New hip? Follow your surgeon's precautions first.

Standing at the counter (5:00–10:00)

Standing hip-front stretch

2 holds of 20 seconds each side

Same move as before (full instructions on page 33). Cues: Stay tall. Don't arch your back. Small movement, big squeeze. **Breathe:** Breathe slowly.

Easier: Smaller step.

Harder: Reach the arm on the back-leg side up toward the ceiling, to comfortable height.

Calf stretch at the wall

20 seconds each leg

Same move as before (full instructions on page 24). **Cues:** Back heel stays down. Toes point forward. **Breathe:** Breathe slowly through the stretch.

Easier: Smaller step back.

Harder: Bigger step back, or bend the back knee slightly to reach the lower calf.

Standing bird-dog at the counter

6 each side

You need: Kitchen counter. **Set up:** Stand facing the counter, both hands on it.

1. Slide your right foot back and lift it a few inches behind you, toes on or just off the floor.
2. At the same time, if steady, reach your left arm forward along the counter.
3. Return. Switch sides.

Chang's cues: Long back. No arching. Squeeze the buttock of the moving leg. **Breathe:** Breathe out as you reach.

Easier: Leg only; both hands stay on the counter.

Harder: Hold the reach for three seconds.

Counter hip hinge

8 slow hinges

You need: Kitchen counter. **Set up:** Stand arm's length from the counter, hands resting on it, feet hip-width apart, soft knees.

1. Push your hips back, like closing a car door with your bottom.
2. Let your chest tip forward with a long, flat back. Hands slide along the counter.
3. Stop when you feel a stretch in the back of your thighs.
4. Squeeze your buttocks to stand tall again.

Chang's cues: Hips go back, not down. Long back, chest proud. No rounding. Stand up by squeezing your buttocks. **Breathe:** Breathe out as you stand tall.

Easier: Smaller movement, hands stay on the counter the whole time.

Harder: Pick up a jug of water from the seat of a chair using the same hinge, and set it back down. Later: from a lower surface, like a step.

Safety: Osteoporosis or a past spine fracture: this is the safe way to bend. Keep your back long; never round forward under load.

Close (10:00–11:00)

4-6 breathing

1 minute

Same move as before (full instructions on page 33). Cues: Belly rises first. Soft shoulders. **Breathe:** In for 4, out for 6. That is about 6 breaths a minute. No holds.

Safety: If you feel light-headed, go back to normal breathing.

Easier: In for 3, out for 4.

Harder: Continue for 5 minutes.

Chang Yin

Notice how you feel getting out of bed tomorrow. Same time tomorrow.

Today's habit: After-dinner walk. Walk 10 minutes after your biggest meal. Even 2 to 5 minutes of light walking after eating lowers the blood-sugar rise compared with sitting (E23). Indoors is fine: kitchen to front door, and back.

☐ Session done ☐ Habit done Effort (1 easy to 5 hard):
____ Any pain above 3/10? ☐ No ☐ Yes

Day 5: Legs, a little faster

Time: about 11 minutes. **You need:** Chair against a wall, bottom stair with a handrail (or a sturdy step), kitchen counter.

Chang Yin

Today we stand up faster.

Not sloppy. Faster. Fast-intent training, lifting quick and lowering slow, gives older adults functional gains similar to or a little better than regular training (E39). Getting up quick is what saves you when you trip.

Stop rule. Stop if you feel chest pain or pressure, dizziness, unusual shortness of breath, or sharp pain. Chest pain, face drooping, arm weakness or trouble speaking? Call 911 now. Pain rule: Mild discomfort up to 3 out of 10 is okay if it settles by tomorrow. Sharper, or worse the next day: back off and ask your doctor or physical therapist.

At a glance

Minutes	Part	Moves
0:00–2:00	Warm-up	Standing march at the counter (1 minute); Chair sit-to-stand (1 easy set of 5)

Minutes	Part	Moves
2:00–9:30	Main block	Power stand (fast up, slow down) (3 sets of 5. Rest 45 seconds between sets); Step-ups on the bottom stair (2 sets of 6 each leg); Counter hip hinge (2 sets of 8); Side steps along the counter (2 sets of 10 steps each way); Heel raises at the counter (1 set of 12)
9:30–11:00	Cool-down	Calf stretch at the wall (20 seconds each leg); Cyclic sighing (1 minute)

Warm-up (0:00–2:00)

Standing march at the counter

1 minute

Same move as before (full instructions on page 23). Cues: Stand tall. Knee up, foot down softly. **Breathe:** Breathe steadily. You should be able to talk.

Easier: Sit and do the seated march.

Harder: Lift knees higher, march faster, or let one hand hover above the counter.

Chair sit-to-stand

1 easy set of 5

Same move as before (full instructions on page 21). Cues: Nose over toes. Knees point the same way as your toes. Don't let them cave inward. Sit down slow. Two or three seconds. **Breathe:** Breathe out as you stand. Breathe in as you sit.

Easier: Put a firm cushion or folded blanket on the seat to make it higher, or push up with your hands on your thighs or on chair arms.

Harder: Arms crossed, lower for a slow count of three, and pause one second just above the seat before touching down. Later: hold a gallon jug of water at your chest.

Main block (2:00–9:30)

Power stand (fast up, slow down)

3 sets of 5. Rest 45 seconds between sets.

You need: Sturdy chair against a wall. **Set up:** Sturdy chair with no wheels and no arms (unless the easier version uses arms), pushed back against a wall so it cannot slide.

1. Sit tall, feet flat, arms crossed or hands on thighs.
2. Stand up as quickly as you safely can. Think 'pop up'.
3. Stand tall for one second.
4. Lower slowly for a count of three.

Chang's cues: Fast up, slow down. Quick is not the same as sloppy. Knees stay over toes. **Breathe:** Breathe out hard as you stand.

Easier: Normal-speed sit-to-stand with hands on your thighs.

Harder: Hold a jug of water at your chest, or pause one second just above the seat on the way down.

Safety: Stop the fast part if you feel dizzy when you stand. Go back to normal speed.

Step-ups on the bottom stair

2 sets of 6 each leg

You need: Bottom stair with a handrail. **Set up:** Stand facing the bottom step of your stairs. Hold the handrail the whole time. Good light, no rug, shoes on.

1. Step up with your right foot, placing the whole foot on the step.
2. Push through that heel to bring your left foot up. Stand tall.
3. Step down with the same leg you started with, slowly.
4. Do all reps leading with one leg, then switch.

Chang's cues: Whole foot on the step, not just the toes. Push through the heel of the top foot. Rail hand stays on the rail. **Breathe:** Breathe out as you step up.

Easier: Toe taps: tap the step with one foot, then the other, holding the rail. Or use a lower step, like a sturdy doorway threshold.

Harder: Step up without pushing off the bottom foot. Later: carry a light bag of groceries in the hand that isn't on the rail.

Safety: If stairs make you dizzy or you have fallen on stairs, do toe taps only until you've practiced balance for two weeks.

Counter hip hinge

2 sets of 8

Same move as before (full instructions on page 51). Cues: Hips go back, not down. Long back, chest proud. No rounding. Stand up by squeezing your buttocks. **Breathe:** Breathe out as you stand tall.

Safety: Osteoporosis or a past spine fracture: this is the safe way to bend. Keep your back long; never round forward under load.

Easier: Smaller movement, hands stay on the counter the whole time.

Harder: Pick up a jug of water from the seat of a chair using the same hinge, and set it back down. Later: from a lower surface, like a step.

Side steps along the counter

2 sets of 10 steps each way

Same move as before (full instructions on page 31). Cues: Stay tall. Small, controlled steps. Feet don't cross. **Breathe:** Breathe normally. Talk or count out loud.

Easier: Fewer steps, slower.

Harder: Bend your knees slightly and stay low, or tie a resistance loop above your knees.

Heel raises at the counter

1 set of 12

Same move as before (full instructions on page 22). **Cues:** Tall like someone is pulling a string on top of your head. Weight over the big toe and second toe, not rolling out. **Breathe:** Breathe out as you rise.

Safety: Calf cramps are common at first. Stretch and walk them out. One calf that is swollen, red and painful is not a cramp: that is for your doctor, today.

Easier: Sit in your chair and raise your heels from there, or hold the counter with both hands and rise only halfway.

Harder: One hand on the counter. Later: one leg at a time, with both hands on the counter.

Cool-down (9:30–11:00)

Calf stretch at the wall

20 seconds each leg

Same move as before (full instructions on page 24). **Cues:** Back heel stays down. Toes point forward. **Breathe:** Breathe slowly through the stretch.

Easier: Smaller step back.

Harder: Bigger step back, or bend the back knee slightly to reach the lower calf.

Cyclic sighing

1 minute

Same move as before (full instructions on page 25). Cues: Two sniffs in, one long sigh out. No holding at any point. **Breathe:** The out-breath is always longer than the in-breath. No breath holds.

Safety: If you feel light-headed, return to normal breathing.

Easier: Just breathe slowly through your nose, making the out-breath a little longer.

Harder: Build up to 5 minutes.

Chang Yin

Fast up, slow down. You did it. Same time tomorrow.

Today's habit: Commercial-break stands. Every time you get up from the sofa or the table, do it twice: stand, sit, then stand. Two extra stands, a dozen times a day, adds up.

☐ Session done ☐ Habit done Effort (1 easy to 5 hard):

____ Any pain above 3/10? ☐ No ☐ Yes

Day 6: Breathe, flow, walk

Time: about 12 minutes. **You need:** Space to stand, a chair behind you, kitchen counter, walking shoes.

Chang Yin

Today is softer. Not easier. Softer.

Slow breathing, then five pieces of a gentle qigong set called baduanjin, then a short walk. We teach tai chi and qigong as exercise: slow weight shifts, stepping and turning. Balance is one of the most trainable things there is. We do a small taste today.

Stop rule. Stop if you feel chest pain or pressure, dizziness, unusual shortness of breath, or sharp pain. Chest pain, face drooping, arm weakness or trouble speaking? Call 911 now. Pain rule: Mild discomfort up to 3 out of 10 is okay if it settles by tomorrow. Sharper, or worse the next day: back off and ask your doctor or physical therapist.

At a glance

Minutes	Part	Moves
0:00–2:00	Breath	Cyclic sighing (2 minutes)

Minutes	Part	Moves
2:00–6:30	Baduanjin, five pieces	Holding up the sky (baduanjin piece 1) (4 slow reps); Drawing the bow (baduanjin piece 2, high stance) (3 each side); Separating heaven and earth (baduanjin piece 3) (3 each side); Looking back (baduanjin piece 4, gentle) (3 each side); Punching with a steady gaze (baduanjin piece 7, high stance) (4 each side)
6:30–7:30	Tai chi	Cloud hands (tai chi, simplified) (1 minute, slow)
7:30–11:00	Walk	Walk and talk (3 minutes 30 seconds, easy to brisk)
11:00–12:00	Close	4-6 breathing (1 minute, seated)

Breath (0:00–2:00)

Cyclic sighing

2 minutes

Same move as before (full instructions on page 25). Cues: Two sniffs in, one long sigh out. No holding at any point. **Breathe:** The out-breath is always longer than the in-breath. No breath holds.

Safety: If you feel light-headed, return to normal breathing.

Easier: Just breathe slowly through your nose, making the out-breath a little longer.

Harder: Build up to 5 minutes.

Baduanjin, five pieces (2:00–6:30)

Holding up the sky (baduanjin piece 1)

4 slow reps

You need: Space to stand; chair behind you. **Set up:** Stand tall, feet hip-width apart, knees soft, chair behind you. Interlace your fingers in front of your belly, palms up.

1. Breathe in and lift your hands slowly up the front of your body, turning palms to the ceiling as they pass your face.
2. Press gently upward, only to a comfortable height.
3. Breathe out and float your arms down to the sides.

Chang's cues: Slow, like moving through warm water. Look straight ahead.

Breathe: Breathe in as the hands rise, out as they come down.

Easier: Sit and do it seated, or lift only to shoulder height.

Harder: Rise onto your toes a little at the top, holding the chair back with one hand if needed.

Safety: Shoulder pain or a known rotator cuff problem? Only lift to a comfortable height, or skip it.

Drawing the bow (baduanjin piece 2, high stance)

3 each side

You need: Space to stand; chair beside you. **Set up:** Stand with feet a little wider than your hips, knees slightly bent. A high stance, not a deep squat.

1. Bring both hands in front of your chest.
2. Push your left hand out to the left like holding a bow, while your right elbow pulls back like drawing the string. Eyes follow the left hand.
3. Return to center. Switch sides.

Chang's cues: Knees over toes. Grow tall through the top of the head.

Breathe: Breathe out as you draw the bow, in as you return.

Easier: Sit tall in your chair and do the arm movement only.

Harder: Bend your knees a little more (never below comfortable).

Separating heaven and earth (baduanjin piece 3)

3 each side

You need: Space to stand; chair behind you. **Set up:** Stand tall, feet hip-width apart, hands in front of your belly.

1. Lift your right hand up past your face and press the palm toward the ceiling, to comfortable height.
2. At the same time, press your left palm down beside your hip.
3. Bring both hands back to center. Switch.

Chang's cues: One hand reaches up, one presses down. Long spine.

Breathe: Breathe in as you separate, out as you return.

Easier: Seated, or lift only to shoulder height.

Harder: Slower, with a longer out-breath.

Safety: Shoulder pain or a known rotator cuff problem? Only lift to a comfortable height, or skip it.

Looking back (baduanjin piece 4, gentle)

3 each side

You need: Space to stand or sit. **Set up:** Stand or sit tall, arms relaxed by your sides.

1. Turn your palms outward and slowly turn your head to look over your right shoulder, only as far as comfortable.
2. Return to center. Switch.

Chang's cues: Chin level. Shoulders stay down. **Breathe:** Breathe out as you turn, in as you return.

Easier: Turn your eyes more than your head.

Harder: Hold the turn for one slow breath.

Safety: Nods and half-turns only. Never full neck circles. Osteoporosis? Turn gently and only as far as comfortable. Never force the twist.

Punching with a steady gaze (baduanjin piece 7, high stance)

4 each side

You need: Space to stand; chair beside you. **Set up:** Stand with feet a little wider than hips, knees soft, loose fists at your waist, palms up.

1. Slowly punch your right fist forward, turning it palm down.
2. Open the hand, grip the air, and pull the fist back to the waist.
3. Switch sides.

Chang's cues: Slow punch. This is not boxing. Eyes steady and focused.

Breathe: Breathe out as you punch.

Easier: Seated, same arm movement.

Harder: Bend your knees a little more, keeping them over your toes.

Tai chi (6:30–7:30)

Cloud hands (tai chi, simplified)

1 minute, slow

You need: Space to stand; counter or chair nearby. **Set up:** Stand with feet a little wider than hips, knees soft, near the counter.

1. Shift your weight to the right foot while your right hand floats across in front of your face and your left hand floats low in front of your belly.
2. Shift to the left foot and let the hands switch, like wiping a big window slowly.
3. Keep shifting side to side.

Chang's cues: Move from your waist and legs, not just your arms. In tai chi we call it sinking: really, it's lowering your center of mass. Soft knees.
Breathe: Breathe slowly; no need to match the breath to the moves.

Easier: Keep one hand on the counter and move the other hand only.

Harder: Add a small side step with each shift, next to the counter.

Walk (7:30–11:00)

Walk and talk

3 minutes 30 seconds, easy to brisk

You need: Shoes, a safe route (hallway, mall, sidewalk). **Set up:** Good shoes. A route with no ice, good light, and benches or rails if possible. Cane or walker if you use one.

1. Walk at an easy pace for 2 minutes to warm up.
2. Pick up the pace to 'can talk, can't sing' for the middle minutes.
3. Slow down for the last 2 minutes.

Chang's cues: Tall. Eyes ahead, not at your feet. Arms swing naturally. Heel, then toe. **Breathe:** Breathe easy and steady. The talk test: you should be able to speak in short sentences.

Easier: Walk indoors: kitchen to front door and back, touching the counter as you pass. Or march seated.

Harder: Add 1-minute faster stretches, or a gentle hill.

Safety: Chest pain, pressure, or unusual breathlessness: stop, sit, and call your doctor. If it doesn't settle in a few minutes, call 911.

Close (11:00–12:00)

4-6 breathing

1 minute, seated

Same move as before (full instructions on page 33). Cues: Belly rises first. Soft shoulders. **Breathe:** In for 4, out for 6. That is about 6 breaths a minute. No holds.

Safety: If you feel light-headed, go back to normal breathing.

Easier: In for 3, out for 4.

Harder: Continue for 5 minutes.

Chang Yin

Five minutes of slow breathing before bed tonight. That's the habit. Same time tomorrow.

Today's habit: Five-minute wind-down. Before bed, five minutes of cyclic sighing: two breaths in through the nose, one long breath out through the mouth. In a one-month study of 111 adults, it improved daily mood more than mindfulness meditation (E18). No breath holds.

☐ Session done ☐ Habit done Effort (1 easy to 5 hard):
____ Any pain above 3/10? ☐ No ☐ Yes

Day 7: Put it together, then retest

Time: about 8 minutes, then the retest. **You need:** Everything from this week.

Chang Yin

Last day. Short session, then we measure.

Measure twice. Lift once. Warm up with me, then rest five minutes and redo your six tests.

Stop rule. Stop if you feel chest pain or pressure, dizziness, unusual shortness of breath, or sharp pain. Chest pain, face drooping, arm weakness or trouble speaking? Call 911 now. Pain rule: Mild discomfort up to 3 out of 10 is okay if it settles by tomorrow. Sharper, or worse the next day: back off and ask your doctor or physical therapist.

At a glance

Minutes	Part	Moves
0:00–1:30	Warm-up	Standing march at the counter (45 seconds); Weight shifts (4 shifts each direction)
1:30–8:00	Full-body round	Chair sit-to-stand (1 set of 8); Wall push-up (1 set of 8); Heel-to-toe stand at the counter (2 holds of 10 seconds each side); One-arm jug row at the counter (1 set of 8 each arm); Heel raises at the counter (1 set of 10); Cyclic sighing (30 seconds)

Warm-up (0:00–1:30)

Standing march at the counter

45 seconds

Same move as before (full instructions on page 23). **Cues:** Stand tall. Knee up, foot down softly. **Breathe:** Breathe steadily. You should be able to talk.

Easier: Sit and do the seated march.

Harder: Lift knees higher, march faster, or let one hand hover above the counter.

Weight shifts

4 shifts each direction

Same move as before (full instructions on page 28). **Cues:** Move from your ankles and hips, like a tree swaying. Feet stay flat on the floor.

Breathe: Breathe slowly.

Easier: Do it seated: shift your weight side to side on the chair.

Harder: Make the shifts bigger, or let one hand hover.

Full-body round (1:30–8:00)

Chair sit-to-stand

1 set of 8

Same move as before (full instructions on page 21). **Cues:** Nose over toes. Knees point the same way as your toes. Don't let them cave inward. Sit down slow. Two or three seconds. **Breathe:** Breathe out as you stand. Breathe in as you sit.

Easier: Put a firm cushion or folded blanket on the seat to make it higher, or push up with your hands on your thighs or on chair arms.

Harder: Arms crossed, lower for a slow count of three, and pause one second just above the seat before touching down. Later: hold a gallon jug of water at your chest.

Wall push-up

1 set of 8

Same move as before (full instructions on page 38). **Cues:** Body in one straight line, head to heels. Chest to the wall, not your chin. **Breathe:** Breathe out as you push away.

Easier: Stand closer to the wall and move only halfway.

Harder: Step your feet farther back. When 15 are easy, move to counter push-ups.

Heel-to-toe stand at the counter

2 holds of 10 seconds each side

Same move as before (full instructions on page 29). **Cues:** Eyes on something still at eye level. Soft knees. Tall head. **Breathe:** Breathe slowly and steadily.

Safety: Balance drills always start with support. Hands hover, they don't wander away from the counter.

Easier: Half-step stance: front foot's heel beside the big toe of the back foot. Both hands on the counter.

Harder: Lift one hand to hover just above the counter. Later: both hands hovering.

One-arm jug row at the counter

1 set of 8 each arm

Same move as before (full instructions on page 38). **Cues:** Pull with your elbow, not your hand. Long back. Don't twist. **Breathe:** Breathe out as you pull.

Safety: Osteoporosis: keep the hinge small and the back long.

Easier: Stand upright and pull a towel looped around a sturdy door handle (see Towel row), or use a lighter jug.

Harder: Full gallon jug (about 8 pounds), or a bag with a few cans. Pause for one second at the top.

Heel raises at the counter

1 set of 10

Same move as before (full instructions on page 22). **Cues:** Tall like someone is pulling a string on top of your head. Weight over the big toe and second toe, not rolling out. **Breathe:** Breathe out as you rise.

Safety: Calf cramps are common at first. Stretch and walk them out. One calf that is swollen, red and painful is not a cramp: that is for your doctor, today.

Easier: Sit in your chair and raise your heels from there, or hold the counter with both hands and rise only halfway.

Harder: One hand on the counter. Later: one leg at a time, with both hands on the counter.

Cyclic sighing

30 seconds

Same move as before (full instructions on page 25). **Cues:** Two sniffs in, one long sigh out. No holding at any point. **Breathe:** The out-breath is always longer than the in-breath. No breath holds.

Safety: If you feel light-headed, return to normal breathing.

Easier: Just breathe slowly through your nose, making the out-breath a little longer.

Harder: Build up to 5 minutes.

Chang Yin

You did seven days. Now look at your numbers. Then decide what's next.

Today's habit: Call one person. Call someone today and tell them what you did this week. Stronger social ties were linked with a 50% higher likelihood of survival across 148 studies (E37). A link, not a promise. But call anyway.

☐ Session done ☐ Habit done Effort (1 easy to 5 hard):
____ Any pain above 3/10? ☐ No ☐ Yes

Your seven habit cards

One small habit a day, attached to something you already do. Cut these out and stick today's card on the fridge. By Day 7 you'll have tried all seven. Keep the two you liked best.

Day 1: Sun's protein breakfast. Put 25 to 30 grams of protein in your breakfast (E28). Two eggs plus a cup of soy milk or a small Greek yogurt gets most people there. Kidney disease? Ask your doctor for your number first.

☐ Done today

Day 2: Toothbrush balance. While you brush your teeth, stand heel-to-toe with your free hand on the sink or a grab bar. Switch feet halfway. Two minutes, twice a day, is 28 minutes of balance practice a week without trying. Home balance practice is a core part of the Otago programme (E13).

☐ Done today

Day 3: Two-bag carry. Every time you bring groceries in, carry two lighter bags, one in each hand, standing tall. It's the carry exercise with a purpose. Pick bags up with a hinge, long back, never a rounded one.

☐ Done today

Day 4: After-dinner walk. Walk 10 minutes after your biggest meal. Even 2 to 5 minutes of light walking after eating lowers the blood-sugar rise compared with sitting (E23). Indoors is fine: kitchen to front door, and back.

☐ Done today

Day 5: Commercial-break stands. Every time you get up from the sofa or the table, do it twice: stand, sit, then stand. Two extra stands, a dozen times a day, adds up.

☐ Done today

Day 6: Five-minute wind-down. Before bed, five minutes of cyclic sighing: two breaths in through the nose, one long breath out through the mouth. In a one-month study of 111 adults, it improved daily mood more than mindfulness meditation (E18). No breath holds.

☐ Done today

Day 7: Call one person. Call someone today and tell them what you did this week. Stronger social ties were linked with a 50% higher likelihood of survival across 148 studies (E37). A link, not a promise. But call anyway.

☐ Done today

Day 7: the retest

Rest five minutes after today's session. Have some water. Then redo all six tests exactly the way you did on Day 0: same chair, same shoes, same weight for the curl, same helper if you had one. Write the numbers in the Day 7 column of your score sheet, and work out your Strength Age again.

Sun Yoon

Sun Yoon. His wife. Also AI. Here's the honest version: in seven days, most of the change is your body getting better at the movement. Your brain learns the move. That's real. Building actual muscle takes weeks of training two or three times a week (E01, E02). So if your number went up one or two, good. If it didn't move, also fine. You started. Seven out of ten. Because I love you.

What counts as progress this week:

- One or two more chair stands. The chair stand is the number that usually moves first.
- A position further on the balance test, or more seconds on the next one.
- The Timed Up and Go a second or two quicker, walking at your normal pace.
- Getting up from the sofa without thinking about it.
- You did seven sessions. That's the habit, and the habit is the program.

If a number went down: tired, sore, a bad night's sleep, a different time of day. Retest in a week. If something hurts more than 3 out of 10 and isn't settling, or anything on the red-flag list appears, stop and call your doctor.

What's next: Strong Years

Chang Yin

Seven days is a start. Strength is a habit. Habits need tomorrow.

Muscle is a savings account. Deposit now. You'll need it later.

You can repeat this week on your own. Here's how: do Days 1, 3 and 5 as your strength days, two or three times a week (that's what the research uses, E01, E02), Day 2 and Day 4 in between, and Day 6 on the weekend. When a move feels easy for all the reps, switch to its Harder version.

Or keep going with us in **Strong Years**, the membership this Reset is the front door to:

- **A new 8 to 12 minute session every day** with Chang Yin. The week rotates: Monday strength, Tuesday mobility, Wednesday balance, Thursday strength, Friday breath and qigong, Saturday walk-and-talk, Sunday rest and stretch.
- **Four levels, so it always fits:** Rebuild (all chair-based), Steady, Strong and Iron. A "sore knee / sore back / low energy today" button swaps in a gentler version.
- **Six 12-week programs** to finish, one at a time: Strong at 70, Back Strong, Balance and Steady Feet, Gut Reset with Sun Yoon, Grip and Hands, and Walk Stronger.
- **Sun Yoon's Kitchen every Sunday:** three recipes, a grocery list and one kitchen remedy graded honestly.
- **Your Strength Age retest every month**, charted, so you can see the line move.

- **A daily message** from Chang by email at the time you choose, **Ask Chang / Ask Sun** AI chat that remembers your level if you allow it, and a **real human coach** on a live Q&A every Wednesday.

What it costs, plainly. Your Strong Years founding membership is **\$30 a month**, and it stays \$30 for as long as you stay subscribed. It renews every month **until you cancel**, and every membership comes with a **14-day money-back guarantee** on the first monthly charge. We email you a reminder before every renewal. Cancel online anytime, no phone call, from your account page (strongyears.com/account), or email us and a person handles it within one business day. If you bought the books on their own, nothing renews; you can add the membership later.

Whatever you decide: keep your score sheet. Retest in 30 days. The number is yours.

Your 7-day tracker

Stick this on the fridge. Tick it every day. Grace rule: miss a day, do the next day's session tomorrow. Don't double up.

Day	Date	Session ☐	Habit ☐	Effort 1-5	Pain above 3/10? Where?
Day 1: Legs first					
Day 2: Steady feet					
Day 3: Arms, back and grip					
Day 4: Loosen up					
Day 5: Legs, a little faster					
Day 6: Breathe, flow, walk					
Day 7: Put it together, then retest					

My best number this week: _____

The move I'm proudest of: _____

The habit I'm keeping: _____

Who I told: _____

Where the numbers come from

Every number and health statement in this book traces to one of the entries below. They are the same evidence IDs our whole team uses (EVIDENCE.md). Grade A means a Cochrane review, a large meta-analysis or a guideline body. B means a single good trial or a small meta-analysis. C means an observational study: it shows a link, not cause and effect. D means lab, physiology or label data.

E01 (grade A). Progressive resistance training in older adults: 121 RCTs, 6,700 people, 2–3×/week. Large effect on strength (SMD 0.84), moderate-to-large effect on getting out of a chair (SMD –0.94), gait speed +0.08 m/s, less osteoarthritis pain. Most side effects were minor soreness or joint pain.

Source: Liu & Latham, Cochrane 2009 — https://www.cochrane.org/evidence/CD002759_progressive-resistance-strength-training-improving-physical-function-older-adults

E02 (grade A). NSCA position statement: resistance training is safe and effective for older adults, including frail people. Progress gradually. Train 2–3×/week, 1–3 sets of 6–12 reps at 70–85% 1RM once adapted. Include power (fast lifting intent) and balance.

Source: Fragala et al., J Strength Cond Res 2019 — https://journals.lww.com/nsca-jscr/fulltext/2019/08000/resistance_training_for_older_adults__position.1.aspx

E03 (grade B (small)). Ten frail nursing-home residents aged ~90, 8 weeks of high-intensity leg training: strength +174%, thigh muscle area up about 9%, walking speed improved. Muscle still adapts in the 90s.

Source: Fiatarone et al., JAMA 1990 — <https://pubmed.ncbi.nlm.nih.gov/2342214/>

E04 (grade A (review)). Around age 75, muscle mass drops about 0.6–1% per year while strength drops 2.5–4% per year. Strength declines 2–5× faster than mass.

Source: Mitchell et al., *Frontiers in Physiology* 2012 — <https://www.frontiersin.org/journals/physiology/articles/10.3389/fphys.2012.00260/full>

E05 (grade B). 10 days of bed rest in healthy adults (mean age 67) cost about 0.95 kg of leg lean mass, with muscle protein synthesis down about 30%. Younger adults lost under 0.4 kg over 28 days.

Source: Kortebein et al., *JAMA* 2007 — <https://news.uams.edu/2007/04/25/extended-bed-rest-accelerates-muscle-deterioration-in-older-adults-uams-researchers-report-in-jama/>

E06 (grade A (consensus)). Sarcopenia screening cut-offs (EWGSOP2): grip under 27 kg for men or under 16 kg for women, or 5 chair rises taking more than 15 s, signals "probable sarcopenia".

Source: Cruz-Jentoft et al., *Age Ageing* 2019 — <https://www.esceo.org/sites/esceo/files/pdf/2019%20Age%20Ageing%20EWGSOP2.pdf>

E07 (grade C). PURE study: 142,861 adults in 17 countries. Every 5 kg lower grip strength was linked with 16% higher all-cause mortality and 17% higher CV mortality. Grip predicted mortality better than systolic blood pressure.

Source: Leong et al., *Lancet* 2015 — [https://www.thelancet.com/journals/lancet/article/PIIS0140-6736\(14\)62000-6/abstract](https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(14)62000-6/abstract) ; summary <https://www.consultant360.com/exclusives/could-grip-strength-predict-all-cause-mortality-risk>

E09 (grade C). 10-second one-leg stance (free foot resting on the back of the other calf, arms at sides, 3 tries). 1,702 people aged 51–75. Failing it was linked with 84% higher all-cause mortality over about 7 years. Failure rates: about 18% at 61–65, about 37% at 66–70, over 50% at 71–75.

Source: Araújo et al., *BJSM* 2022 — <https://www.bristol.ac.uk/news/2022/june/tne-second-one-legged-stance.html>

E10 (grade C). Gait speed and survival: 34,485 adults 65+. 0.8 m/s is around median life expectancy. ≥ 1.0 m/s was linked with longer-than-expected survival. Under 0.6 m/s means higher risk.

Source: Studenski et al., JAMA 2011 — <https://www.sciencedaily.com/releases/2011/01/110104161621.htm>

E11 (grade A (tool)). CDC STEADI 30-second chair stand (17-inch chair, arms crossed), used in the CDC's older-adult screening. Below-average scores: men 60–64 <14, 65–69 <12, 70–74 <12, 75–79 <11, 80–84 <10, 85–89 <8, 90–94 <7. Women 60–64 <12, 65–69 <11, 70–74 <10, 75–79 <10, 80–84 <9, 85–89 <8, 90–94 <4. If arms are used to stand, the score is 0.

Source: CDC STEADI — <https://www.cdc.gov/steady/media/pdfs/STEADI-Assessment-30Sec-508.pdf>

E13 (grade A–). Otago Exercise Programme: a home programme of leg strengthening, balance exercises and walking designed for older adults, studied in 7 trials with 1,503 people (mean age 81.6). We use its exercise types and progressions; we make no outcome claim from it.

Source: Thomas et al., Age Ageing 2010 — <https://www.ncbi.nlm.nih.gov/books/NBK78898/>

E15 (grade B+). Tai chi vs physical therapy for knee osteoarthritis: 204 adults. Similar improvements in pain and function at 12 weeks, sustained to 52 weeks. Tai chi group had more improvement in depression and quality of life.

Source: Wang et al., Ann Intern Med 2016 — <https://www.nccih.nih.gov/research/research-results/study-shows-tai-chi-and-physical-therapy-were-equally-helpful-for-knee-osteoarthritis>

E17 (grade A–). Baduanjin/qigong in older adults: meta-analyses report better sleep quality and fewer depressive symptoms (moderate certainty, heterogeneous trials).

Source: Frontiers Public Health 2026 meta-analysis — <https://pmc.ncbi.nlm.nih.gov/articles/PMC13501274/> ; qigong & sleep meta — <https://pmc.ncbi.nlm.nih.gov/articles/PMC12757222/>

E18 (grade B). 111 adults, 5 min/day for 1 month. Cyclic sighing (two inhales through the nose, long slow exhale through the mouth) produced the largest daily improvement in positive mood, more than mindfulness meditation. All breathing groups improved.

Source: Balban et al., Cell Reports Medicine 2023 — <https://stanmed.stanford.edu/cyclic-sighing-stress-relief/>

E19 (grade A– (meta)). Voluntary slow breathing (around 6 breaths/min) increases vagally-mediated heart-rate variability during and after practice.

Source: Laborde et al., Neurosci Biobehav Rev 2022 — <https://www.sciencedirect.com/science/article/abs/pii/S0149763422002007>

E20 (grade A). Exercise and resting BP: 270 RCTs, 15,827 people. Isometric training lowered BP the most (–8.24/–4.00 mmHg). Wall squat ranked top for systolic. Aerobic –4.49/–2.53, dynamic resistance –4.55/–3.04. These are average changes over weeks of training, not instant.

Source: Edwards et al., BJSM 2023 — <https://bmjgroup.com/static-isometric-exercise-such-as-wall-sits-best-for-lowering-blood-pressure/>

E22 (grade C). Daily steps: in adults 60+, mortality benefit plateaued at about 6,000–8,000 steps/day. The top quartiles had 40–53% lower risk than the lowest (about 3,500/day).

Source: Paluch et al., Lancet Public Health 2022 — <https://www.sciencedaily.com/releases/2022/03/220303112207.htm>

E23 (grade A– (meta)). Light walking after meals (even 2–5 minutes) lowers the post-meal glucose rise compared with sitting.

Source: Buffey et al., Sports Med 2022 — <https://pubmed.ncbi.nlm.nih.gov/35147898/>

E28 (grade A (consensus)). PROT-AGE: adults over 65 should get 1.0–1.2 g protein/kg/day, and 1.2–1.5 g/kg if active or ill (unless there is severe kidney disease). Aim for 25–30 g per meal with about 2.5–2.8 g leucine.

Source: Bauer et al., JAMDA 2013 — <https://pubmed.ncbi.nlm.nih.gov/23867520/>

E30 (grade A). Exercise for chronic low back pain: 249 trials, 24,486 people. Pain about 15 points better (0–100 scale) at 3 months vs no treatment or usual care. Function improved modestly.

Source: Hayden et al., Cochrane 2021 — https://www.cochrane.org/evidence/CD009790_exercise-treatment-chronic-low-back-pain

E37 (grade C (meta of cohorts)). 148 studies, 308,849 people. Stronger social relationships were linked with a 50% higher likelihood of survival (OR 1.50), comparable to well-known risk factors like smoking.

Source: Holt-Lunstad et al., PLoS Med 2010 — <https://journals.plos.org/plosmedicine/article?id=10.1371%2Fjournal.pmed.1000316>

E39 (grade A–). High-velocity "power" training (lift with fast intent, lower slowly) gives functional gains similar to or slightly better than traditional training in older adults, and it is safe when supervised and progressed.

Source: J Physiotherapy 2023 systematic review — <https://www.sciencedirect.com/science/article/pii/S183695532300053X> ; Eur Rev Aging Phys Act 2022 — <https://link.springer.com/article/10.1186/s11556-022-00297-x>

E40 (grade A (consensus)). Osteoporosis exercise guidance: do resistance and balance training. Avoid rapid, repeated or loaded spinal flexion (sit-ups, toe-touches with weight) and forceful twisting. Flexion-exercise programs were linked with more vertebral fractures.

Source: Giangregorio et al., "Too Fit To Fracture", Osteoporos Int 2014 — <https://link.springer.com/10.1007/s00198-014-2881-4> ; Sinaki & Mikkelsen, Arch Phys Med Rehabil 1984

E41 (grade B (known literature)). Pain-monitoring model: during and after exercise, pain up to about 3/10 (we use a conservative threshold) is acceptable if it settles by the next morning and isn't trending up week to week.

Source: Thomeé, Phys Ther 1997; Silbernagel et al., Am J Sports Med 2007

E42 (grade A (statement)). AHA scientific statement (2023): resistance training is recommended for adults with and without cardiovascular disease. It lowers blood pressure and improves glucose control and function. Avoid breath-holding (Valsalva). People with CVD should get individualized clearance.

Source: Paluch et al., Circulation 2024 — <https://www.ahajournals.org/doi/10.1161/CIR.0000000000001189>

E43 (grade A (guideline)). Pre-exercise screening (ACSM 2015): anyone with symptoms of cardiovascular, metabolic or renal disease (chest pain, unusual breathlessness, fainting) should get medical clearance before starting. Otherwise, light-to-moderate exercise can begin and progress.

Source: Riebe et al., Med Sci Sports Exerc 2015 (known literature)

E46 (grade A– (meta, known literature)). Orthostatic hypotension (a blood-pressure drop on standing) is common in older adults. A meta-analysis found it in about 22% of community-dwelling older people. The risk is higher with some blood-pressure and other medicines. Practical rule: sit, pump your ankles, then stand slowly while holding something.

Source: Saedon et al., J Gerontol A Biol Sci Med Sci 2020 (prevalence of orthostatic hypotension, systematic review & meta-analysis)

E47 (grade D (physiology)). The calf muscle pump: calf contraction compresses the deep veins and, with the venous valves, pushes blood back toward the heart. It's standard physiology. Walking and heel raises activate it. A red flag is one swollen, red, painful calf (possible clot), which needs a doctor, not exercise.

Source: Standard physiology texts; e.g. Recek, Int J Angiol 2013 "Calf pump activity influencing venous hemodynamics in the lower extremity" (known literature)

E49 (grade A (tool/norms)). Senior Fitness Test (Rikli & Jones): 7,183 US community-dwelling adults aged 60–94 from 267 sites in 21 states. "Normal range" = the middle 50% (25th–75th percentile) for each 5-year age band and sex. 30-s chair stand, 30-s arm curl (5 lb women / 8 lb men), 2-minute step test (right-knee lifts to midway between kneecap and hip bone), chair sit-and-reach (inches, + past the toes / – short of them). Example normal ranges, women 70–74: chair stand 10–15, arm curl 12–17, step test 68–101, sit-and-reach –1.0 to +4.0 in; men 70–74: 12–17, 14–21, 80–110, –3.5 to +2.5 in. Full tables in products/strength_reset.md. It's a fitness comparison, not a diagnosis.

Source: Rikli & Jones, Senior Fitness Test Manual, 2nd ed. 2013 (primary not opened). Opened: normative tables — <https://integrativephysicaltherapyservices.com/wp-content/uploads/2023/08/Fitness-Testing-for-Seniors-PDF.pdf> ; protocol + sample size — <https://fitnessnorms.com/functional/chair-sit-and-reach/> , <https://fitnessnorms.com/functional/step-test/> , <https://fitnessnorms.com/functional/arm-curl/> . Note: the PDF's women 60–64 sit-and-reach cell reads "–0.5 to +0.5" (typo); we use the fitnessnorms 25th–75th percentiles (–0.6 to +4.8 in).

E50 (grade A (tool)). CDC STEADI 4-Stage Balance Test: feet side by side, semi-tandem (instep beside big toe), tandem (heel to toe), one leg; up to 10 s each, stop at the first stage not held. The CDC uses the tandem stand as a screening checkpoint: under 10 s is worth discussing with a clinician.

Source: CDC STEADI — opened <https://www.cdc.gov/steady/media/pdfs/STEADI-Assessment-4Stage-508.pdf>

E51 (grade A (tool)). CDC STEADI Timed Up and Go: stand from a chair, walk 10 feet (3 m) at normal pace, turn, walk back, sit. The CDC uses 12 seconds or more as a screening threshold worth discussing with a clinician. Always have someone stand by.

Source: CDC STEADI — opened <https://www.cdc.gov/steady/media/pdfs/STEADI-Assessment-TUG-508.pdf>